STATS 201A Lecture 10

Henry Liev

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Orthogonal Array $(12, 2^{11})$

A	B	
$-$	$-$	
$-$	$+$	$\times 3 = 12$
$+$	$-$	
$+$	$+$	

$$X^\intercal X = \begin{pmatrix} 12 & & \\ & \ddots & \\ & & 12 \end{pmatrix}$$

$$Y = X\beta + \varepsilon, X : 12 \times 12$$

$$\hat{\beta} = (X^\intercal X)^{-1} X^\intercal Y = \frac{1}{12} X^\intercal Y$$

$$\mathrm{Cov}(\hat{\beta}) = \sigma^2 (X^\intercal X)^{-1} = \frac{\sigma^2}{12} \mathbb{I}$$

$k \times k$ Latin square, row column, latin
$OA(k^2, k^3, 2)$
$p = 1 + (2-1) + 8 \cdot (3-1) = 16$
df $= N - p = 18 - 16 = 2$ for residuals, could also include $A \times G$
$OA(18, 6^1 3^6)$
$p = (1 + 16 - 1) + 6 \times (3 - 1) = 18$

$$\mathbb{E}(\hat{A}) - \frac{1}{3} FG = 0 - \frac{1}{3}(-0.4588) = 0.1529$$

$$\mathrm{Cov}(I, ABC) = \frac{4-8}{12} = -\frac{1}{3}$$

$$\begin{aligned} I &= ABC^2 \\ &= A^2 B^2 C \\ A + B + 2C &\equiv 0 \pmod 3 \\ 2A + 2B + C &\equiv 0 \pmod 3 \end{aligned}$$